

Research Article

Quad Port Multipolarized Reconfigurable MIMO Antenna for Sub 6 GHz Applications

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Recent reconfigurable technological advancements for wireless communication systems provide various global solutions. This research work presents a quad port multipolarized switchable multiple input multiple output (MIMO) antenna for sub 6 GHz applications. It covers the frequency range from 3.1 to 5.1 GHz, including the 5G NR band n78 (3.3 to 3.8 GHz) and 5G NR band n79 (4.4 to 5 GHz). The proposed antenna comprises four offset-fed monopole antenna elements with an overall dimension of 60 mm × 65 mm. To achieve circular polarization (CP), a parasitic meandering resonator is integrated with antenna elements using four PIN diodes. The polarization diversity is obtained by controlling the bias states of four PIN diodes. The radiating element −1/−3 offers left hand circular polarization (LHCP), while element −2/−4 procures right hand circular polarization (RHCP) when all diodes are ON. Consequently, the proposed antenna provides linear polarization (LP) under reverse bias conditions. Moreover, the designed antenna acquires a wide axial ratio bandwidth (ARBW) of 36.1%. In addition, the developed MIMO antenna exhibits isolation greater than 15 dB using the common ground plane, and the obtained ECC is less than 0.13. The prototype is fabricated, and the simulated responses are in good correlation with the measured results.

1. Introduction

Reconfigurable MIMO antennas [1] have gained remarkable attention in the modern wireless communication system that minimizes the fading effects over rich multipath environments and improves spectral efficiency using frequency reuse characteristics. Especially, the polarized reconfigurable MIMO antenna (PRMA) emerges as an appropriate solution for indoor environments [2], which resolves connectivity and coverage issues. It is capable of switching between different polarization states such as RHCP, LHCP, and LP. For instance, indoor scenarios such as airports, hospitals, universities, offices, apartments, and shopping malls are equipped with more obstacles, such as concrete walls, furniture, and equipment [3]. Hence, the probability of the occurrence of multipath fading is higher in dense indoor environments [4]. Due to this multipath fading, the signals from the transmitter to reach the receiver may take different

paths at different time instants altering the polarization [5]. Hence, for antenna, the sense of polarization becomes imperative to establish effective communication. The polarization mismatch [6] could be reduced as well as mitigates the signal fading [7] since PRMA can send both vertical and horizontal components. Thus, PRMA must be prominently requisite for indoor wireless communications.

Over the years, many attempts have been reported in [8–12] to achieve circularly polarized (CP) antennas. In [8], a U-shaped slotted patch antenna was published for WLAN application at 5.8 GHz prevailing 2.8% axial ratio bandwidth (ARBW). In [9], a conducting strip shorted to the ground was exploited using PIN diodes to achieve CP with 4.5% ARBW. In [10], an annular slot antenna was designed to resonate at 2.4 GHz and fed with a V-shaped coupling strip to acquire CP with an ARBW of 9%. Moreover, two PIN diodes were deployed to switch between the LP and CP operating states. In [11], a square patch with a loop slot in the

ground plane was presented to produce dual CP modes with an ARBW of less than 7%. To bring in polarization agility, PIN diodes were loaded in the loop slots. In [12], an annular slot ring antenna operating at 2.4 GHz was presented with two perturbation slots to promote orthogonal CP modes with an ARBW of 6.8% using PIN diodes.

Also, various techniques were implemented to achieve different polarizations that still exhibit narrow ARBW of less than 10% and are large in size with a single element. Very few CP MIMO antennas were reported in [13–18] with an effective ARBW greater than 10%. In [13], a quad port CP antenna covers both RHCP and LHCP for the operating frequency of the 5G band from 3.4 to 3.8 GHz with 11.1% ARBW, however, not reconfigurable. In [14], a four element MIMO antenna was reported for the operating frequency 5.8 GHz with an ARBW of 1.58% for WLAN applications. In [15], a dual-port CP MIMO antenna is implemented for WLAN applications with an ARBW of 6.25%. In [16], a DRA-based modified circular-shaped CP MIMO antenna was implemented with an ARBW of less than 7%. In [17], a CP MIMO antenna with polarization diversity was achieved with an ARBW of 15.2% but was not reconfigurable. In [18], the CP MIMO antenna was designed with an ARBW of 15.2% and achieved high gain using a meta-material structure as ground.

Thus, it is evident from the literature developing a wideband CP MIMO antenna with an ARBW greater than 25% is highly challenging, especially for the sub 6 GHz frequency range. Most of the abovementioned works were implemented using techniques, such as shorting vias to cause more radiation loss, dual orthogonal feeding to increase the number of ports, design complexity, and parasitic stubs with low ARBW. But in the proposed work, the parasitic meandering resonator is utilized to bring forth a wide ARBW greater than 35%.

The main contributions of the proposed antenna are as follows:

- (i) The proposed PRMA can dynamically reconfigure the polarization states between LP and CP (both LHCP and RHCP) that can regulate the signal to the desired directions when the radios are placed in dense and varying environments.
- (ii) The proposed MIMO antenna is composed of a mirrored pattern of radiating elements of each other. Therefore, the polarization diversity is controlled by involving less count of PIN diodes.
- (iii) The designed antenna achieves a wide ARBW of 36.1% for the frequency range from 3.4 to 4.9 GHz using a parasitic meandering resonator without increasing the design complexity compared to [13–18].
- (iv) The proposed antenna consists of four antenna elements with the size of $1.25 \lambda_0 \times 1.36 \lambda_0$ (λ_0 -lowest operating frequency of 3.1 GHz) which is more compact than [13–18].
- (v) Also, the proposed antenna achieves a gain of 4.85 dBi greater than [13, 16, 17] and an efficiency of 80%.
- (vi) The developed PRMA is highly suitable for the rich multipath environment since the presented work shows good isolation characteristics and diversity performance.

2. Antenna Design

The unit element is designed over an FR4 substrate with a dielectric constant of 4.4, a loss tangent of 0.023, and a thickness of 1.6 mm. Figure 1 shows the front and rear sides of the designed Polarized Reconfigurable Antenna (PRA) with a size of about $0.27 \lambda_0 \times 0.22 \lambda_0$, where λ_0 corresponds to the lowest operating frequency ($f = 3.1$ GHz). Moreover, the proposed PRA comprises an offset-fed edge tapered rectangular patch (ETRP) with a parasitic meandering resonator that procures degenerated dual orthogonal modes to achieve CP. Also, the unit element is made reconfigurable to switch between two different polarization states, such as LHCP and LP using a PIN diode series BAR50-20V with ON state resistance and inductance of 4.5Ω and 0.6 nH, respectively. During the OFF state, the diode has a very low shunt capacitance of 0.15 pF with reverse resistance of 5 K Ω . For biasing the PIN diode D , two DC blocking capacitors of 20 pF and two RF chokes of 33 nH are utilized to isolate the DC and RF signals, respectively. Therefore, whenever the diode D is forward biased with the voltage 1.2 V the proposed PRA operates in LHCP mode with a 3-dB axial ratio of about 36.1%. On the other hand, when D is unbiased the designed PRA offers LP.

2.1. Evolution of Unit Element. The evolution stages of the ETRP radiator are represented in Figure 2. In stage 1, a simple rectangular monopole patch is symmetrically fed with a transmission line of width 3 mm to match with 50 -ohm impedance. In stage 2, the patch is fed asymmetrically to achieve wide.

Impedance matching that covers the bandwidth ranges from 3.5 to 6 GHz. In stage 3, the feed length is increased from 7.5 mm to 17 mm to radiate from 3.35 GHz. In stage 4, a slot is introduced in the patch to drop the notch around 4 GHz below -10 dB. In stage 5, the top corners of the patch are tapered to have a wide bandwidth that radiates from 3.3 to 5 GHz. In stage 6, a stub is appended in the ground to achieve better impedance matching below -10 dB from 3.1 to 5.1 GHz. Finally, in stage 7, a parasitic meandering resonator is added to the right side of the patch which modifies the current vectors and induces the quadrature phase difference between the degenerated modes over the frequency ranges from 3.4 to 4.9 GHz. Figure 3 depicts the reflection characteristics of the proposed PRA at various stages. Hence, it is observed that the parasitic meandering resonator unaffected the bandwidth of the ETRP radiator, which helps to retain both linear polarization as well as LHCP.

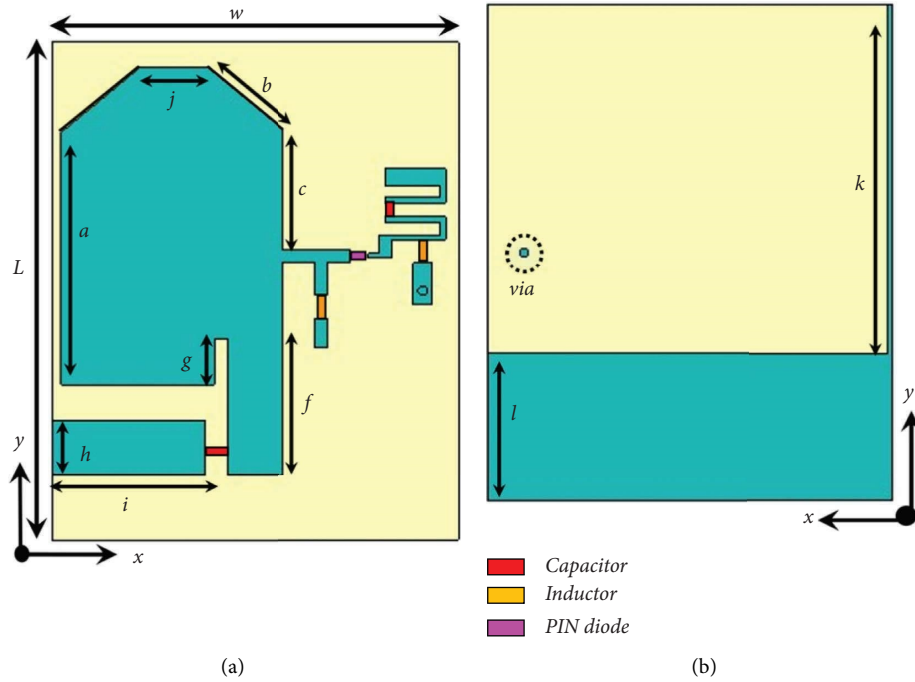


FIGURE 1: Layout of antenna element with dimensions: (a) front view and (b) back view: $L = 27$, $W = 22$, $a = 13$, $b = 5.5$, $c = 7$, $f = 7.5$, $g = 2.5$, $h = 3$, $i = 8.25$, $j = 3.3$, $k = 19$, $l = 8$, and $m = 1.25$. (All dimensions are in mm.).

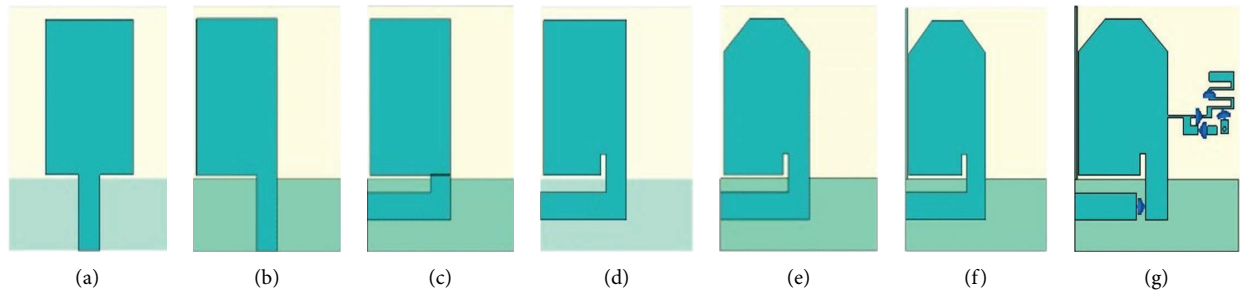


FIGURE 2: Evolution stages of the PRA: (a) stage 1, (b) stage 2, (c) stage 3, (d) stage 4, (e) stage 5, (f) stage 6, and (g) stage 7.

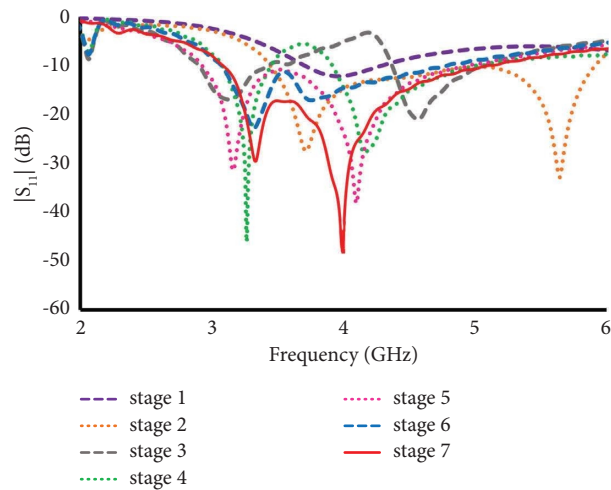


FIGURE 3: Simulated $|S_{11}|$ graphs of the PRA of various stages.

2.2. Circular Polarization Mechanism. CP [19–24] antennas are highly requisite for a rich multipath environment since it diminishes the multipath interferences and are insensitive to antenna orientation between the transmitter and receiver. To attain these dominant characteristics of CP, a parasitic stub is attached to the offset-fed ETRP radiator (Antenna 1) with a length of 11.75 mm. Further to improve the axial ratio, a single turn meandering resonator of length 12 mm is introduced (in antenna 2) reducing the axial ratio below 3 dB at 4.9 GHz. Also, the length of the parasitic meandering resonator is increased to another two turns and now the total length is 24 mm which enlarges the 3 dB ARBW from 3.4–4.9 GHz. Hence, the parasitic meandering resonator induces 90° phase difference and generates dual orthogonal modes. The evolution stages of the proposed CP antenna are represented in Figure 4 and its dimensions are reported in Table 1. However, the effect of varying the number of turns in the parasitic meandering resonator towards reflection characteristics and ARBW is shown in Figures 5(a) and 5(b), respectively. The CP mechanism is explained through the surface current distribution plots. Thus, it is noticed from Figure 6 the dominant current vectors are rotating clockwise along the $+z$ direction, which promotes the LHCP. Additionally, to facilitate multiple polarization vectors a PIN diode D is loaded with biasing circuit in the parasitic meandering resonator. Hence, under the forward bias state, the proposed PRA operates in LHCP mode with a 3-dB axial ratio of about 36.1%.

3. Equivalent Circuit Analysis

The equivalent lumped circuit modelling is implemented by considering the proposed antenna's impedance characteristics [25]. Moreover, the resonance types are predicted based on the real and imaginary curves of the impedance characteristics. The designed equivalent lumped circuit is represented in Figure 7. In Figure 8, the real and imaginary curves of the impedance characteristics as well as the reflection characteristics are illustrated.

4. Polarization Reconfigurable MIMO Antenna (PRMA)

Polarization reconfigurable diversity antennas are the prominent solution to the wireless communication system. Mostly, in varying environments, the multiple polarization vectors are indeed to improve the channel capacity and reduce the multipath interferences. It can dynamically reconfigure between LP, LHCP, and RHCP in varying environments. Thus, a quad port triple polarized diversity antenna is presented to boost the channel capacity and to reduce the signal to noise ratio for indoor scenarios. The proposed PRMA has four radiating elements with a connected ground plane, and its total size is about $1.25 \lambda_0 \times 1.36 \lambda_0$, where λ_0 corresponds to the lowest operating frequency of 3.1 GHz. Moreover, the four radiating elements are arranged in a mirror fashion of each other to procure both LHCP and RHCP. Element 1 and element 3 generate LHCP, whereas an element 2 and 4 produce RHCP

waves. However, four PIN diodes are deployed in the four radiating elements to prevail polarization agility between LP, LHCP, and RHCP for the frequency band of interest (3.1 to 5.1 GHz). When all four diodes are switched off the radiating elements procure linear polarization.

Consequently, the proposed MIMO antenna offers dual sense CP when D_1 , D_2 , D_3 , and D_4 are turned on. Further, when the diode D_1 and D_3 is forward biased and D_2 and D_4 reverse biased the designed MIMO antenna offer simultaneously both LHCP and LP, respectively. Likewise, whenever the diode D_2 and D_3 is forward biased and D_1 and D_4 reverse biased produce RHCP and LP concurrently. Figure 9 portrays the schematic representation of the proposed quad port PRMA.

4.1. Design Process. The ultimate aim of the proposed quad port PRMA is to maintain the wide ARBW and better isolation with the connected ground plane. Moreover, the evolution of the MIMO antenna with the connected ground plane is displayed in Figure 10.

Initially, the four antenna elements are arranged in a mirror pattern with a minimum spacing ($0.18\lambda_0$) to decouple the radiating elements. Further, four connecting stubs were introduced (MIMO antenna 2) to improve the isolation between the four ports, but it deteriorates the ARBW. Besides, in MIMO antenna 3 to sustain better isolation and ARBW the four open-ended stubs were introduced that substantially increase ARBW. Also, two E-shaped stubs were combined with the open-ended stubs. This aids in attaining the isolation and ARBW well below 15 dB and 3 dB, respectively, over the frequency range of 3.3 to 5 GHz. Figure 11 illustrates the evolution stages of S_{11} , S_{21} , and ARBW of the proposed quad port PRMA. Thus, it is observed from Figure 11 there is not much variation in reflection characteristics of desired frequency range from 3.1 to 5.1 GHz while loading the decoupling structure. Hence, the proposed quad port PRMA achieves 36.1% ARBW with a connected ground plane and isolation greater than 15 dB. Figures 12 and 13 illustrate the reflection and isolation characteristics of four radiators of the proposed quad port PRMA.

4.2. Circularly Polarized Mechanism of PRMA. The radiating elements are arranged as mirrored pattern to achieve dual sense CP (LHCP/RHCP). Also, the parasitic meandering resonator is designed to assist the dominant current vectors to rotate in such a way as to have equal magnitude with a 90° phase difference. Hence, to understand this CP mechanism of the proposed MIMO antenna the surface current density plots at frequency 4 GHz are represented in Figure 14.

These current density plots are analysed at four different phases alike $\omega t = 0^\circ, 90^\circ, 180^\circ$, and 270° when all the ports are excited concurrently. It is observed from surface current density plots, the dominant current vectors at 0° and 180° move towards the $+x$ direction (at ports -1 and -2) and $-x$ direction (at ports -3 and -4), respectively. Similarly, at $\omega t = 90^\circ$ and 270° the current vectors move towards the $-y$ direction (at ports -1 and -4) and $+y$ direction (at ports -2

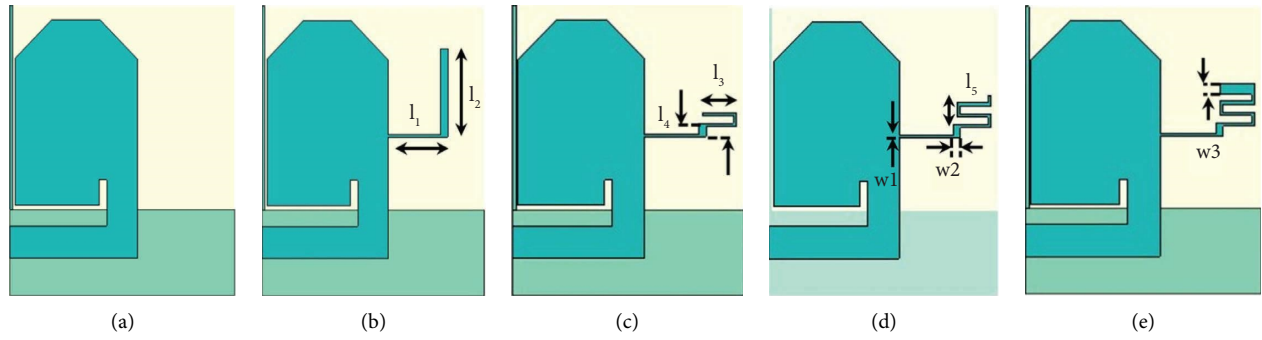


FIGURE 4: Evolution stages of proposed CP antenna: (a) antenna 1, (b) antenna 2, (c) antenna 3, (d) antenna 4, and (e) antenna 5.

TABLE 1: Parameters of parasitic meandering radiator.

Parameters	Values (mm)
l_1	4
l_2	7.75
l_3	3.25
l_4	1
l_5	0.75
w_1	0.75
w_2	0.75
w_3	0.9

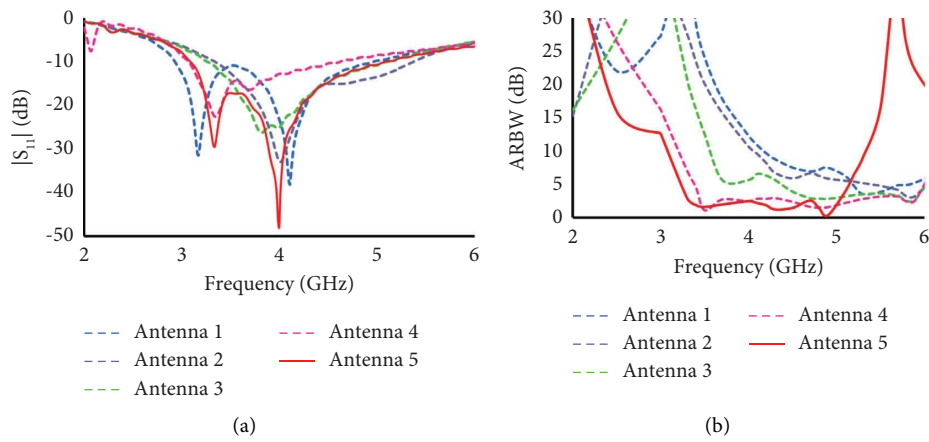


FIGURE 5: Various stages of proposed CP antenna: (a) reflection characteristics and (b) ARBW.

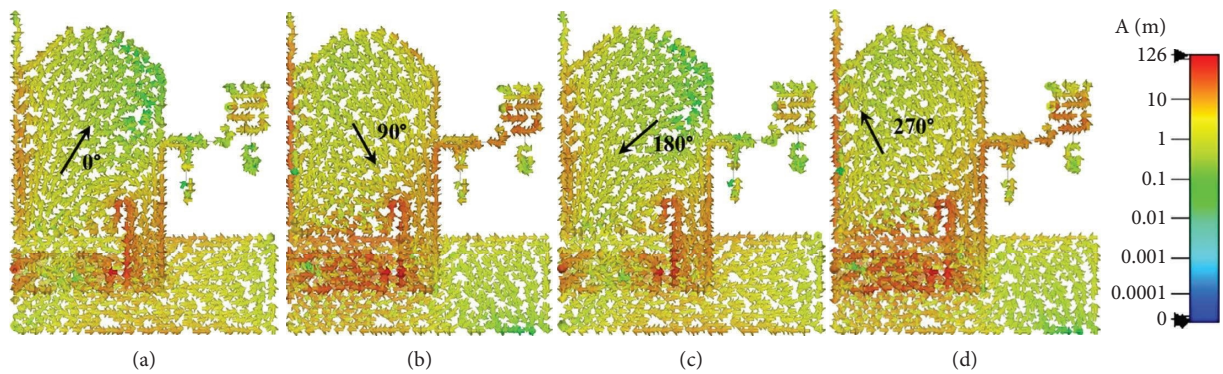


FIGURE 6: Surface current distribution of proposed CP antenna: (a) $\phi = 0^\circ$, (b) $\phi = 90^\circ$, (c) $\phi = 180^\circ$, and (d) $\phi = 270^\circ$.

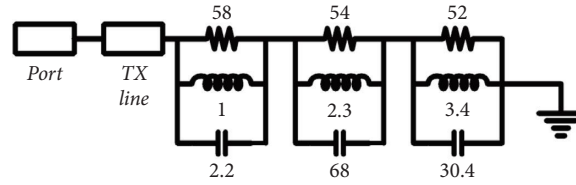


FIGURE 7: Equivalent lumped circuit: resistance, R in Ω ; inductance, L in nH; capacitance, and C in pF.

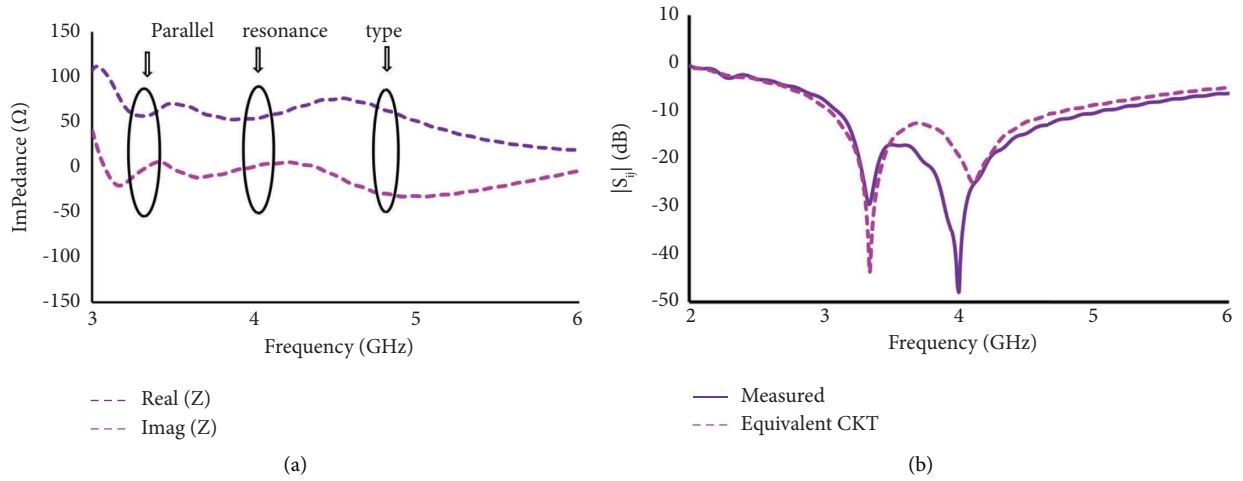


FIGURE 8: (a) Impedance curve and (b) reflection characteristics of the proposed antenna.

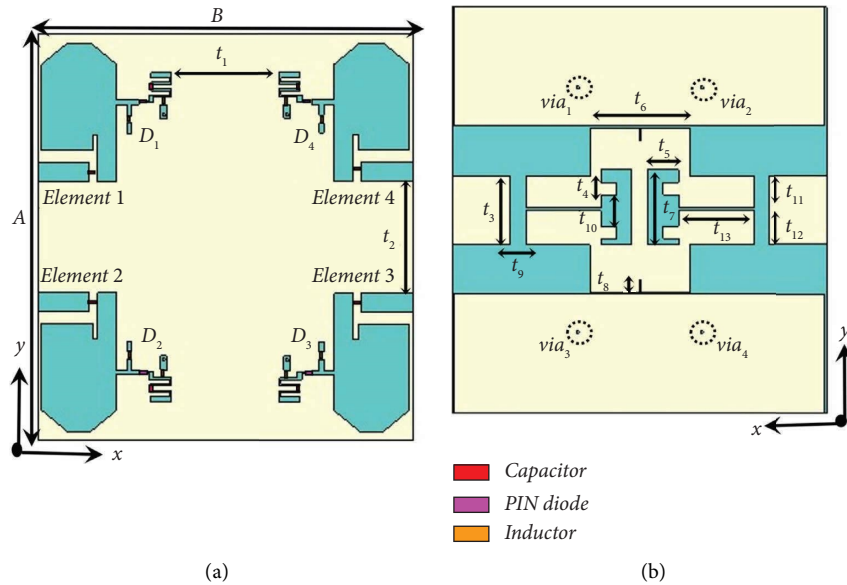


FIGURE 9: Schematic of proposed quad port PRMA: (a) top view and (b) bottom view: $A = 65$, $B = 60$ $t_1 = 0.17 \lambda_0$, $t_2 = 0.18 \lambda_0$, $t_3 = 11$, $t_4 = 2$, $t_5 = 5$, $t_6 = 15.7$, $t_7 = 12$, $t_8 = 2$, $t_9 = 2.5$, $t_{10} = t_{11} = 5$, $t_{12} = 5.5$, and $t_{13} = 12$. (λ_0 -w.r.t lowest operating frequency $f_0 = 3.1$ GHz), (all dimensions are in mm).

and -3), respectively. Thus, the dominant current vectors are equal in magnitude with 90° phase difference aids to generate LHCP waves (at ports -1 and -3) and RHCP waves (at ports -2 and -4).

5. Results and Discussion

The simulation works are carried out using the Finite Difference Time Domain technique tool CST Microwave Studio

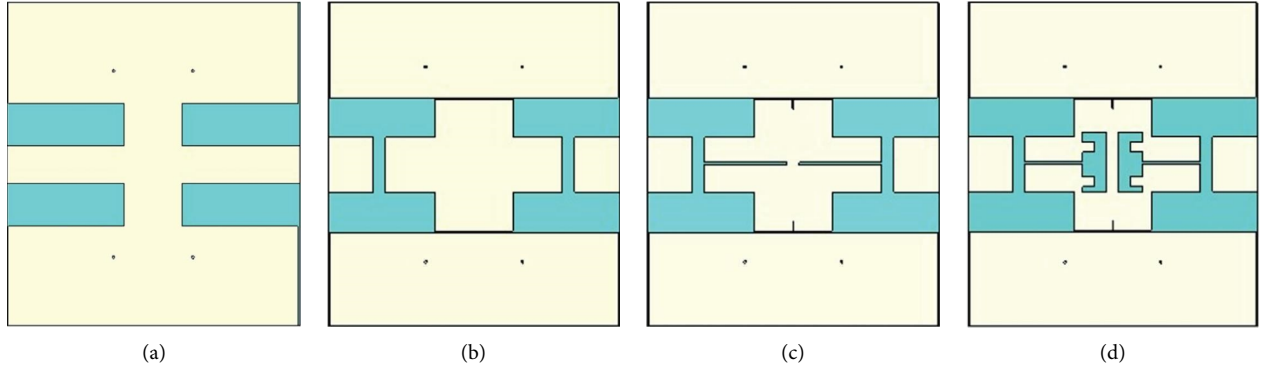


FIGURE 10: Evolution stages of proposed quad port PRMA: (a) MIMO antenna 1, (b) MIMO antenna 2, (c) MIMO antenna 3, and (d) MIMO antenna 4.

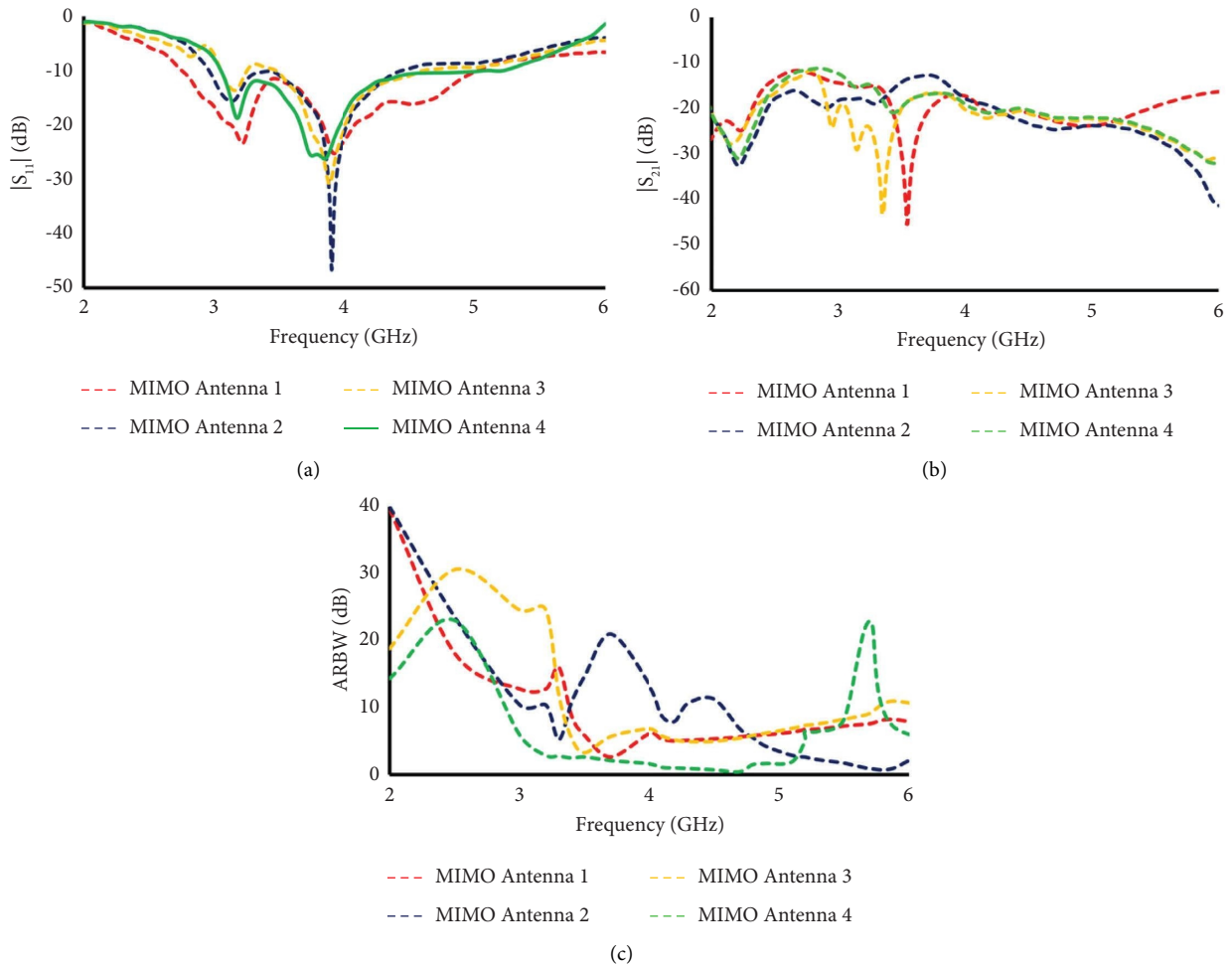


FIGURE 11: Evolution of various stages of quad port PRMA (a) S_{11} , (b) S_{21} , and (c) ARBW.

Suite. The measurements are taken using the vector network analyser (ANRITSU MS2037C), and the pictures are portrayed in Figure 15. The four radiating elements are arranged as mirror images of each other. Due to the identical radiating elements, the reflection characteristics are similar (S_{11} , S_{22} , S_{33} , and S_{44}). Thus, the reflection and radiation

characteristics are evaluated to analyse the antenna performance under different biasing conditions. The proposed antenna achieves the impedance bandwidth of 50% over the desired frequency range from 3.1 to 5.1 GHz for both CP (state 1) and LP (state 2) under forward and reverse biased conditions, respectively. Figure 16 indicates the obtained

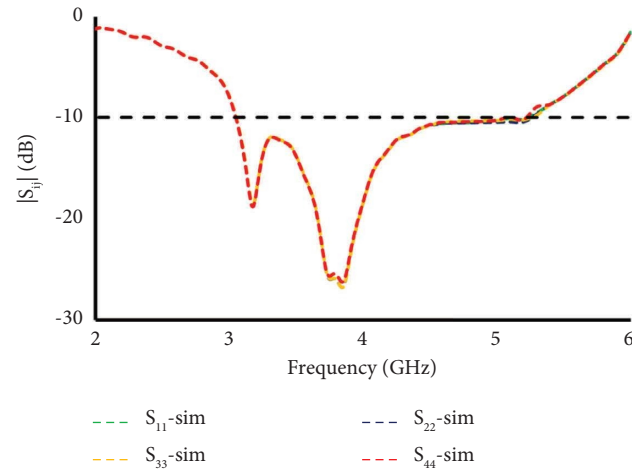


FIGURE 12: Simulated reflection characteristics of quad port PRMA.

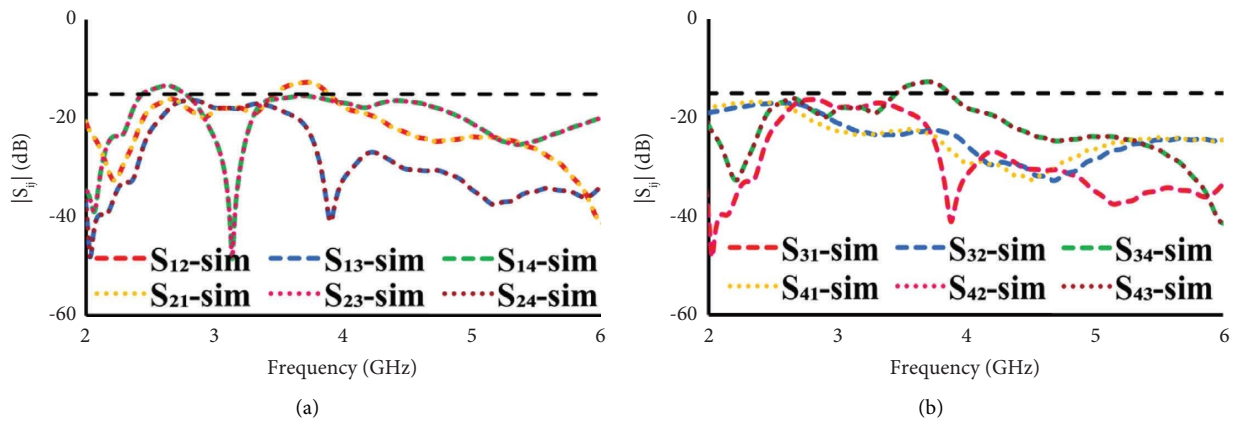


FIGURE 13: Simulated isolation characteristics of quad port PRMA for (a) port 1 and port 2 (b) port 3 and port 4.

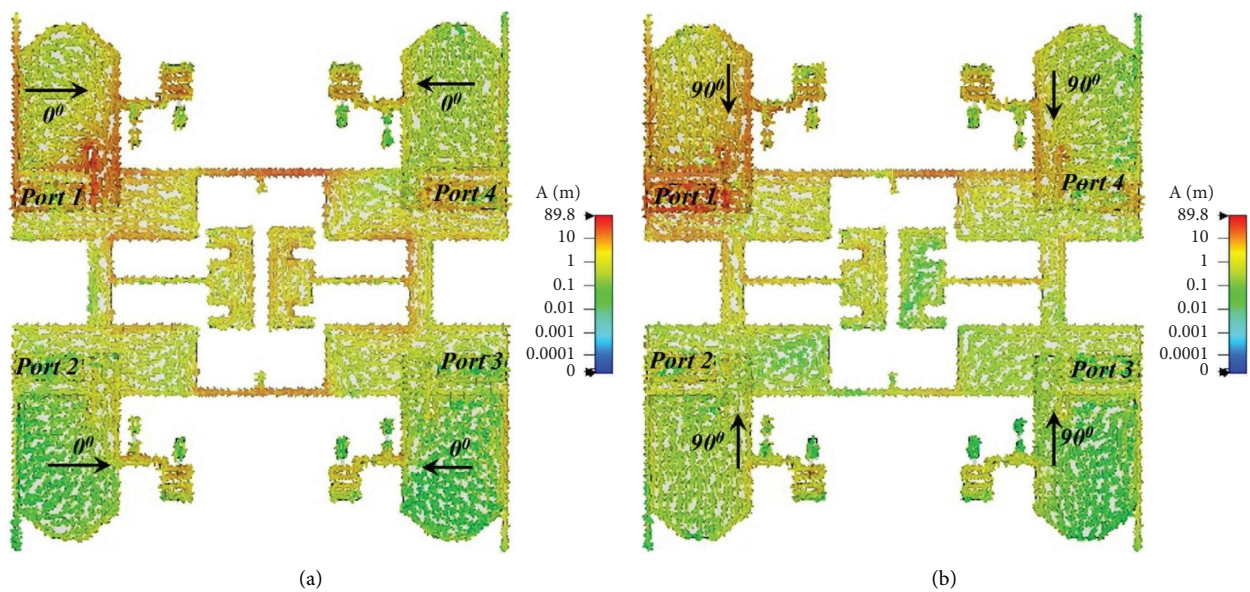


FIGURE 14: Continued.

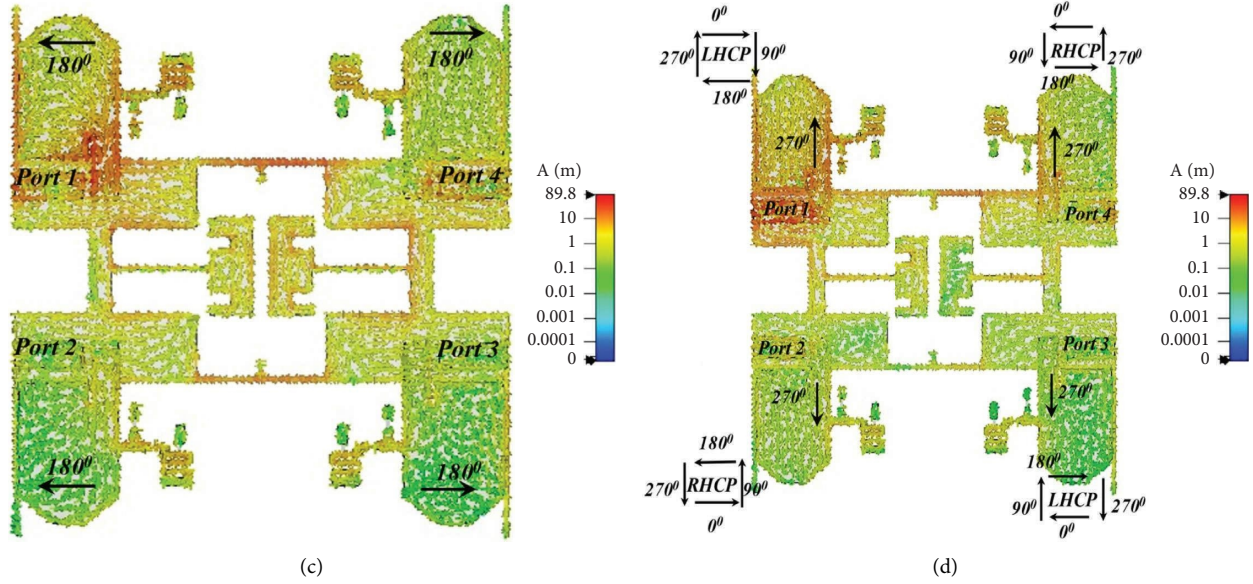


FIGURE 14: Surface current distribution of dual CP quad port MIMO antenna at 4 GHz: (a) $\phi = 0^\circ$, (b) $\phi = 90^\circ$, (c) $\phi = 180^\circ$, and (d) $\phi = 270^\circ$.

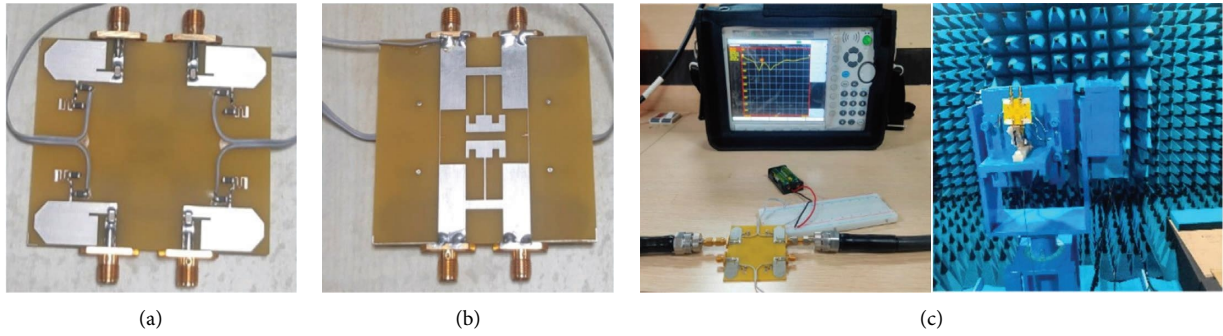


FIGURE 15: Fabricated prototype photographs: (a) front view, (b) rear view, and (c) measurement picture.

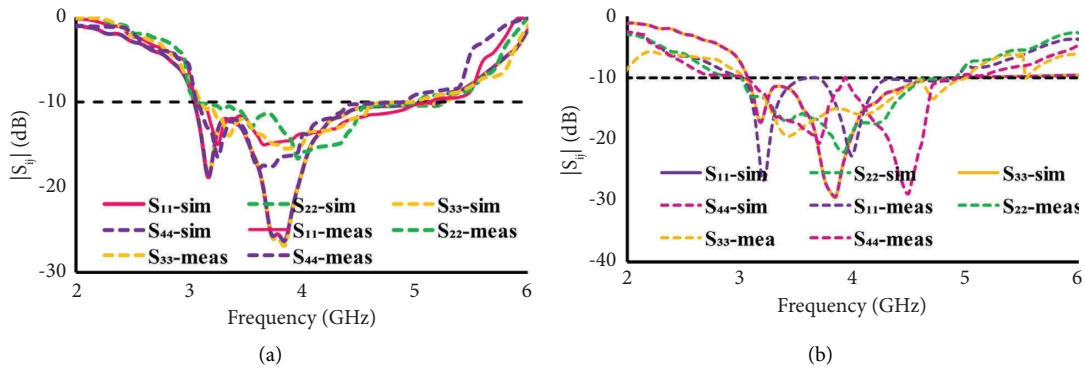


FIGURE 16: Reflection characteristics of proposed quad port MIMO antenna for (a) state 1 and (b) state 2.

reflection coefficient is well below -10 dB over the band of interest for all four ports.

Furthermore, the four radiating elements are separated with the small interelement spacing of $0.17\lambda_0$. Also, the connected ground plane is exhibited to provide an equal

voltage level on the ground surface. This aids to gain the isolation between the radiating elements greater than 15 dB. Figure 17(a) presents the isolation characteristics of the MIMO antenna for ports 1 and 2 with respect to other ports under forward biased conditions. Moreover, Figure 17(b)

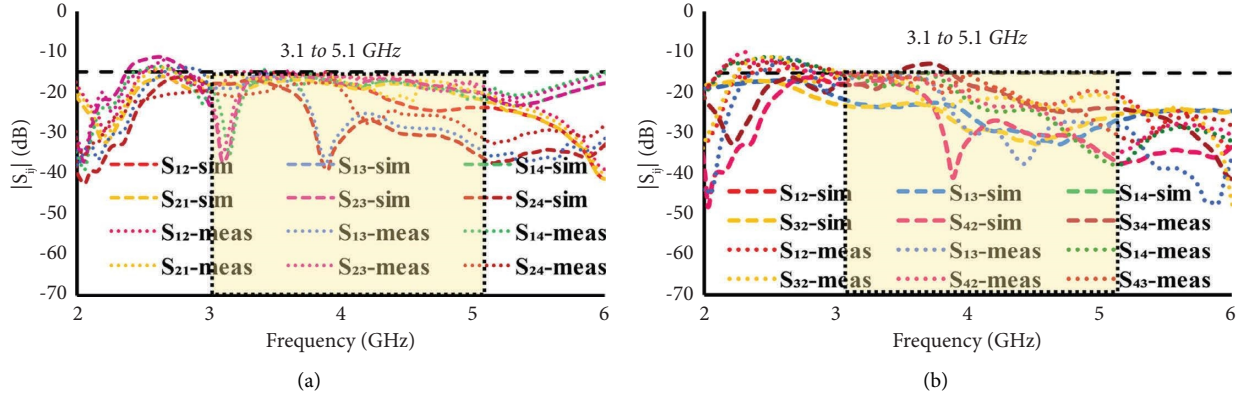


FIGURE 17: Mutual coupling characteristics of ports 1 and 2 for (a) state 1 and (b) state 2 with respect to other ports.

shows the isolation characteristics of ports 1 and 2 with respect to other ports under reverse biased condition. It is noticed that the proposed antenna achieves isolation greater than 15 dB among all ports. However, the small deviation in measured results occurred due to the losses caused by manual soldering of surface-mount devices [25, 26].

Figures 18(a) and 18(b) present the gain and efficiency of the proposed MIMO antenna of both state 1 (CP) and state 2 (LP). In the CP state, the obtained peak gain and efficiency are about 4.85 dBi and 80%, respectively. In the LP state, the acquired peak gain is of 4.5 dBi and 82% efficiency. In addition, the simulated and measured ARBW and gain of the proposed PRMA are illustrated in Figure 18(c). It is evident that the proposed MIMO antenna achieves an ARBW of less than 2.5 dB over the frequency range from 3.3 to 5.1 GHz.

5.1. Radiation Characteristics. The radiation characteristics are obtained under forward biased conditions at frequencies of 3.5 and 4.5 GHz depicted in Figures 19 and 20. It shows that the proposed quad port PRMA generates dual sense CP when the PIN diodes are turned ON. Also, it is observed that ports -1 and 3 generate LHCP signals, whereas ports -2 and 4 produce RHCP signals.

On the other hand, the radiation characteristics at frequencies between 3.5 and 4.5 GHz under a reverse biased condition are illustrated in Figures 21 and 22. At E-plane ($\phi = 90^\circ$), the proposed antenna achieves near dumbbell shape radiation characteristics whereas at H-plane ($\phi = 0^\circ$) near omnidirectional pattern is obtained.

6. Diversity Performance of MIMO Antenna

The diversity characteristics of the proposed quad port PRMA are validated using various metrics such as envelope correlation coefficient (ECC), diversity gain (DG), mean effective gain (MEG), total active reflection coefficient (TARC), and channel capacity loss (CCL).

6.1. Envelope Correlation Coefficient (ECC). The correlation metric (ρ_e) measures the effect of radiated fields of two different antennas when excited simultaneously. Practically,

the ECC should be within 0.3 [27], for the proposed quad port.

PRMA the ECC is computed by considering both scattering parameters and far field characteristics. The mathematical representation is given in the following equations:

$$ECC = \frac{|S_{11}^* S_{12} + S_{21}^* S_{21}|^2}{(1 - |S_{11}|^2 - |S_{21}|^2)(1 - |S_{22}|^2 - |S_{12}|^2)}, \quad (1)$$

$$ECC = \frac{\left| \iint [\vec{F}_i(\theta, \phi) \cdot \vec{F}_j(\theta, \phi)] d\Omega^2 \right|}{\iint |\vec{F}_i(\theta, \phi)|^2 d\Omega \iint |\vec{F}_j(\theta, \phi)|^2 d\Omega}, \quad (2)$$

where F indicates the radiated fields among two different antennas θ and ϕ , and Ω represents the elevation angle, azimuth angle, and solid angle, respectively. Thus, Figures 23(a) and 23(b) present the ECC characteristics of the proposed quad port PRMA under forward and reversed biased conditions, respectively. The ECC is found to be less than 0.2 for the frequency range of 3.1 to 5.1 GHz for both states. Therefore, the lesser correlation between the patterns ensures that the proposed quad port PRMA is an efficient polarization diverse antenna.

6.2. Diversity Gain. Diversity gain (DG) [28] predicts the effect of diversity in the wireless communication system. It can be evaluated using correlation coefficient. The mathematical function for DG is given in the following equation:

$$DG = 10 \sqrt{1 - |\rho_e|^2}. \quad (3)$$

The DG of proposed MIMO antenna is illustrated in Figures 24(a) and 24(b) for both forward biased and reverse biased conditions. It is noticed that the DG is greater than 9 dB for the band of interest 3 to 5 GHz in both cases.

6.3. Mean Effective Gain (MEG). MEG is another significant parameter to measure the diversity characteristics of MIMO antenna. This MEG quantifies the ability to receive electromagnetic power in multipath environment. The

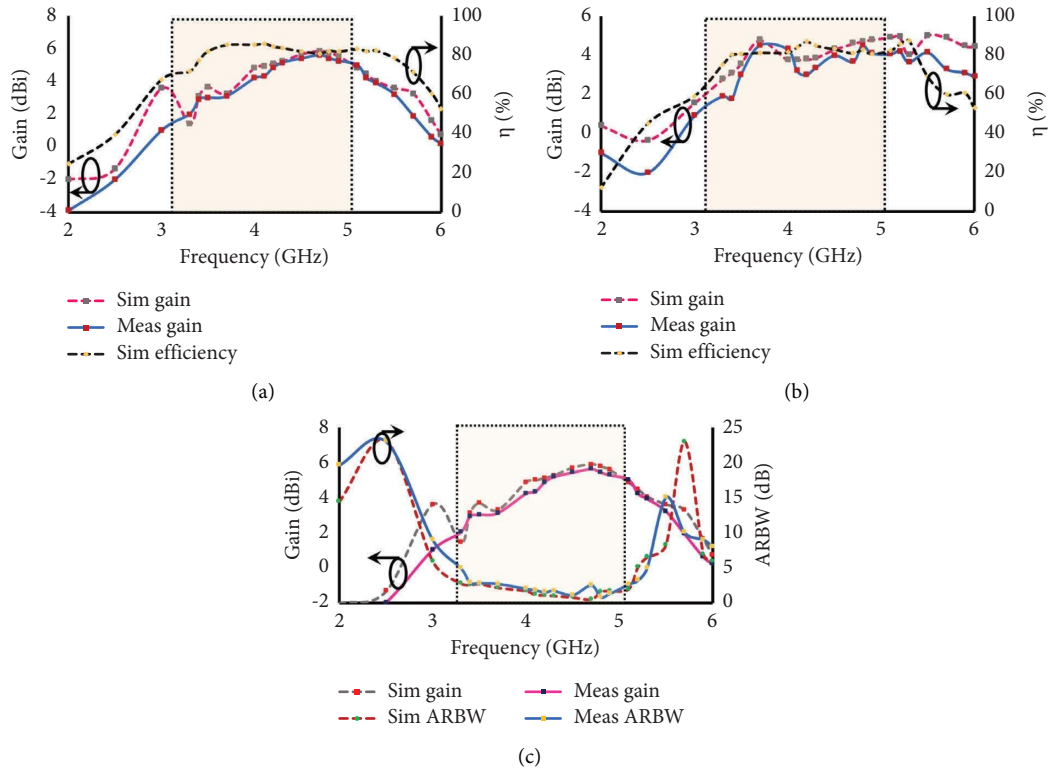


FIGURE 18: Gain and efficiency of (a) state 1, (b) state 2, and (c) measured and simulated gain and ARBW of state 1.

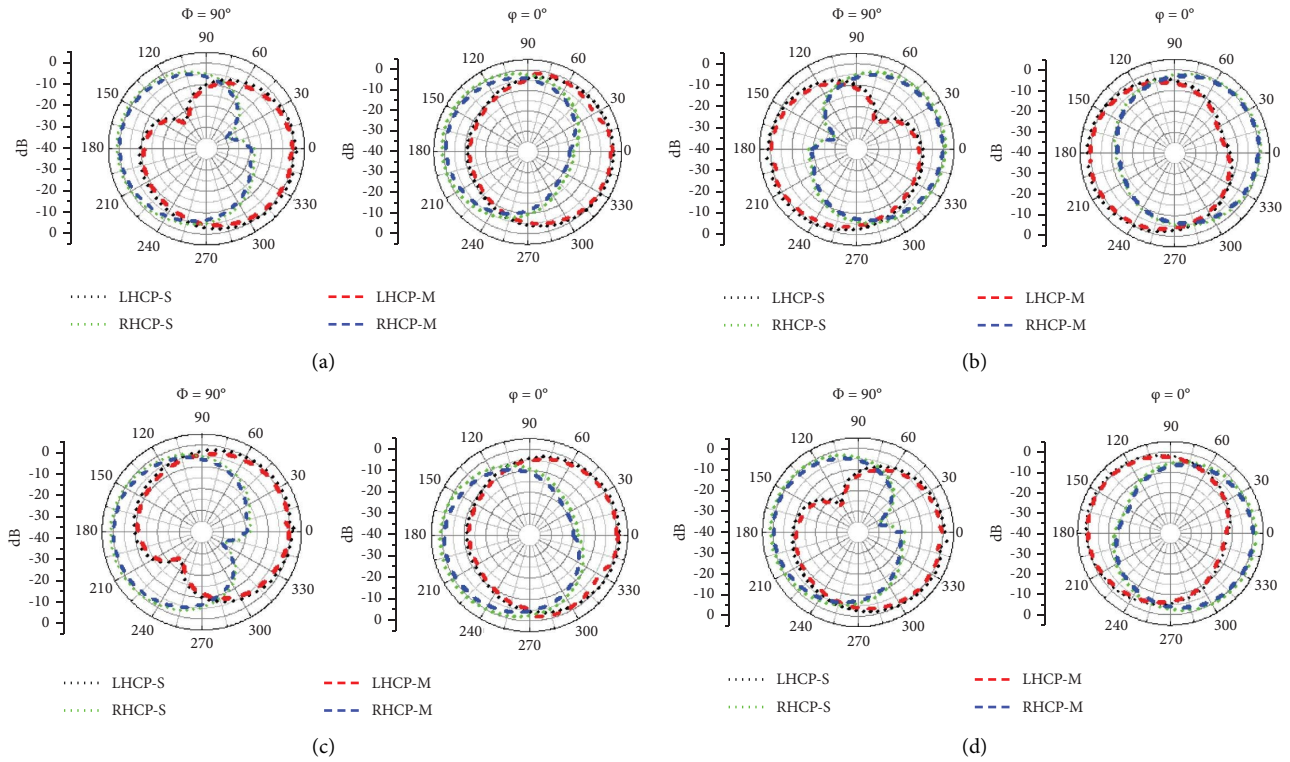


FIGURE 19: Radiation pattern of proposed PRMA for state 1 at 3.5 GHz for (a) port 1, (b) port 2, (c) port 3, and (d) port 4.

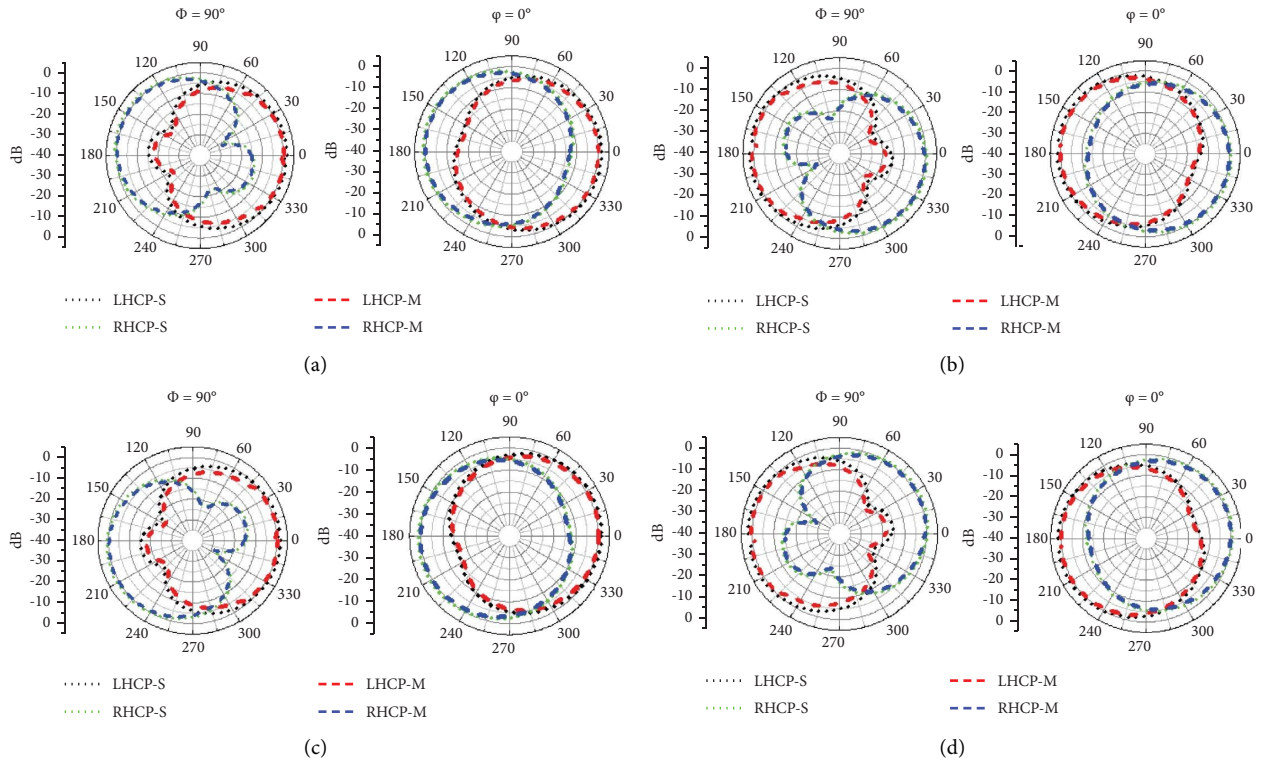


FIGURE 20: Radiation pattern of proposed PRMA for state 1 at 4.5 GHz for (a) port 1, (b) port 2, (c) port 3, and (d) port 4.

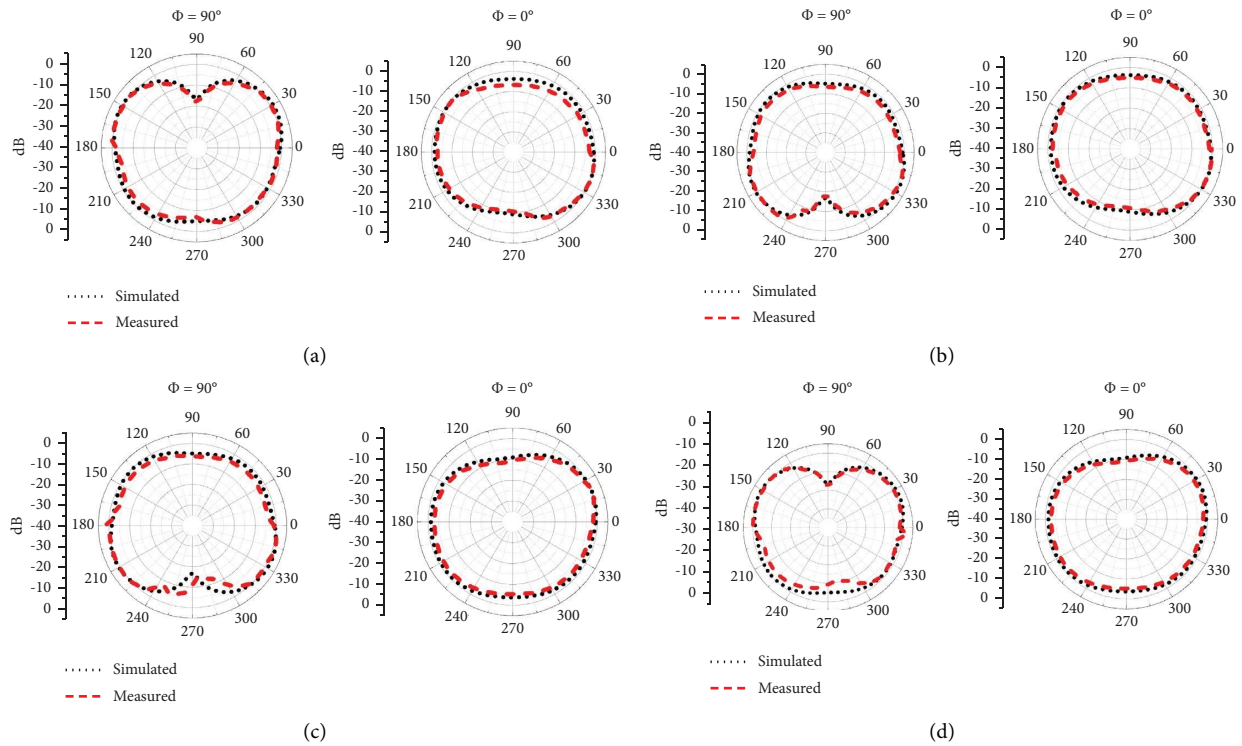


FIGURE 21: Radiation pattern of proposed PRMA for state 2 at 4.5 GHz for (a) port 1, (b) port 2, (c) port 3, and (d) port 4.

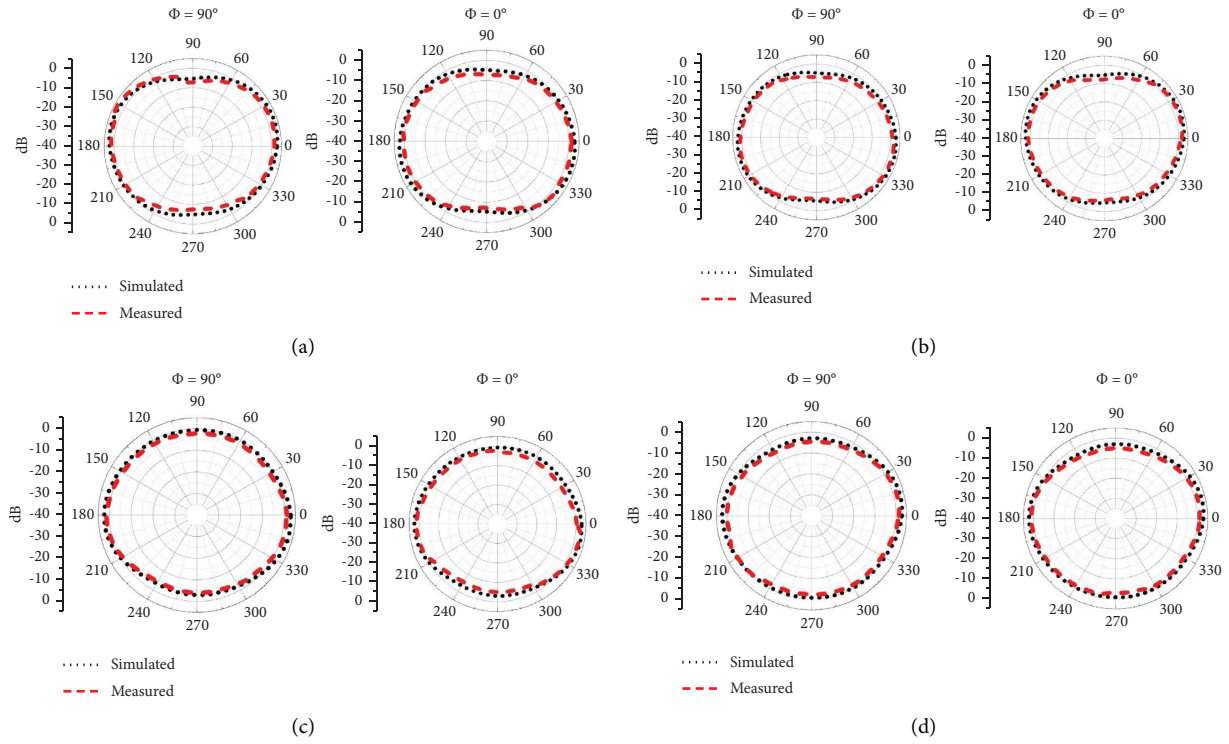


FIGURE 22: Radiation pattern of proposed PRMA for state 2 at 4.5 GHz for (a) port 1, (b) port 2, (c) port 3, and (d) port 4.

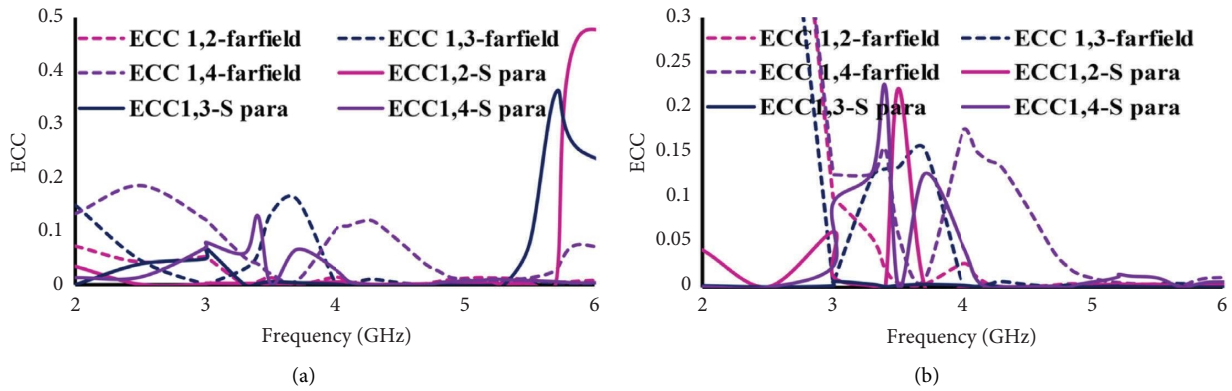


FIGURE 23: ECC characteristics of proposed quad port PRMA for (a) state 1 (CP) and (b) state 2 (LP).

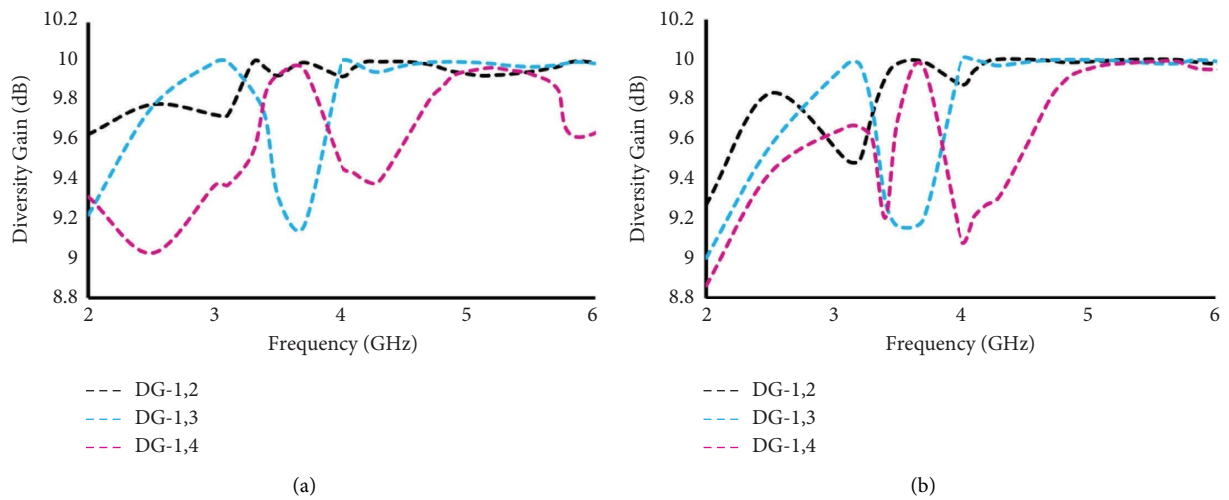


FIGURE 24: Diversity gain of proposed quad port PRMA for (a) state 1 (CP) and (b) state 2 (LP).

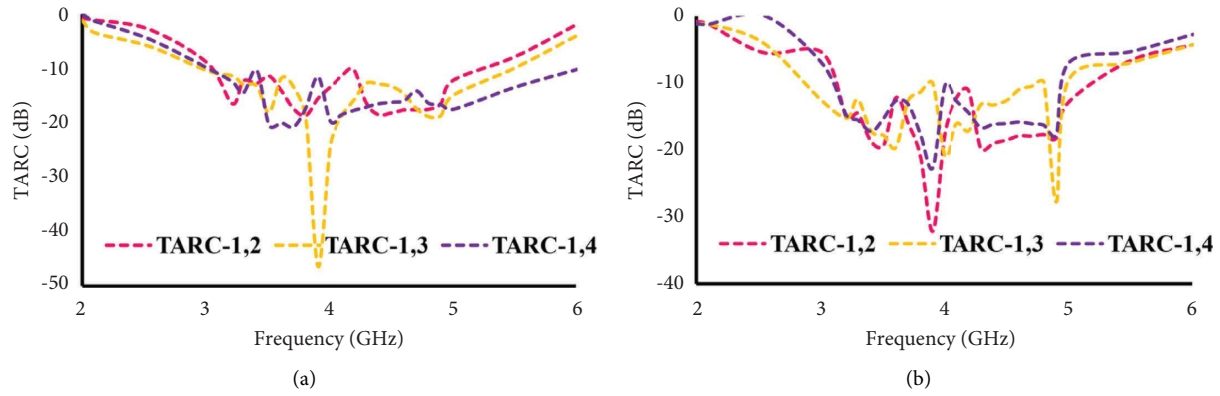


FIGURE 25: TARC characteristics of proposed quad port PRMA for (a) state 1 (CP) and (b) state 2 (LP).

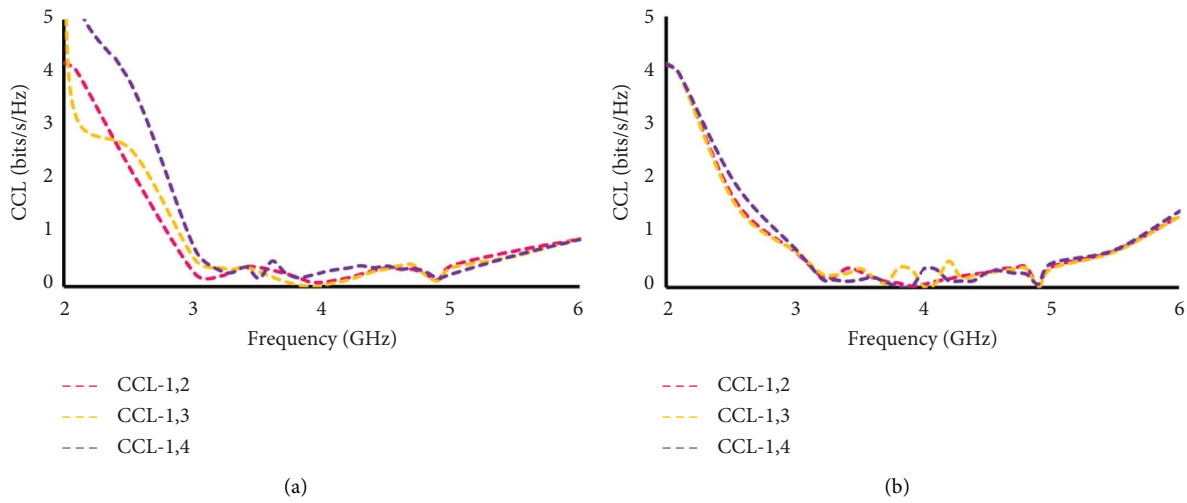


FIGURE 26: CCL characteristics of proposed quad port PRMA for (a) state 1 (CP) and (b) state 2 (LP).

TABLE 2: Diversity metrics of proposed quad port PRMA for port 1 w.r.t other ports.

Frequency (GHz)	Polarization states	Different ports	Diversity metrics				
			ECC (F. F)	DG (dB)	MEG (dB)	TARC (dB)	CCL (bits/s/Hz)
3.5	State 1 (CP)	Ports 1, 2	0.014	9.92	1.005	-12.5	0.349
		Ports 1, 3	0.13	9.3	1.005	-18.11	0.275
		Ports 1, 4	0.02	9.93	1.0	-20.43	0.16
	State 2 (LP)	Ports 1, 2	0.004	9.97	0.995	-19.38	0.028
		Ports 1, 3	0.13	9.16	0.995	-17.9	0.345
		Ports 1, 4	0.05	9.7	1	-15.31	0.15
4.7	State 1 (CP)	Ports 1, 2	0.0004	9.99	0.99	-18.2	0.34
		Ports 1, 3	0.003	9.98	0.995	-13.3	0.33
		Ports 1, 4	0.079	9.59	0.999	-16.2	0.3189
	State 2 (LP)	Ports 1, 2	0.0001	9.99	0.997	-18.5	0.271
		Ports 1, 3	0.002	9.98	0.997	-12.6	0.255
		Ports 1, 4	0.08	9.54	0.999	-16.1	0.294

*F. F-far field.

TABLE 3: Comparison study of proposed work with similar CP MIMO antennas.

R. no	Antenna size (λ_0) (mm ²)	Technique	No. of AEs	f_o (GHz)	ARBW (%)	IBW (%)	PA	Gain (dBi)	Application
[13]	1.42×1.42	Arm length	4	3.4–3.8	11.1	11.1	No	4.5	5G
[14]	1.02×3.66	Two 90° phased rectangular slots	4	5.49–6	1.55	9.27	No	5.34	WLAN
[15]	0.43×0.93	Corner square cut	2	5.23–6.4	6.25	21.4	No	6	WLAN
[16]	1.45×1.82	Modified circular shaped aperture	2	5.25–6.3	11.3	15.1	No	4.7	WLAN
[17]	1.78×1.01	Truncated corner patches	2	5.08–5.9	13.7	15.3	No	5.1	WLAN
[18]	3.18×3.18	Dual feed line	4	11	8.8	15.4	No	13.6	5G
[30]	1.74×0.87	Arc cuts	4	3.48–3.87	4.21	10.9	Yes	3	5G
P.W	1.25×1.36	Parasitic meandering resonator	4	3–5	36.1	50	Yes	4.85	5G

*P.W: proposed work; *: antenna elements; *IBW: impedance bandwidth, *PA: polarization agility.

difference between the MEG [29] of four antennas shows the power imbalance between the antennas. For better performance, the MIMO antenna should possess MEG less than

3 dB. The mathematical representation of MEG is given in the following equation:

$$MEG = \int_0^{2\pi} \int_0^{2\pi} \left[\frac{XPR}{1 + XPR} G_\theta(\theta, \varphi) P_\theta(\theta, \varphi) + \frac{1}{1 + XPR} G_\theta(\theta, \varphi) P_\theta(\theta, \varphi) \right] \sin \theta d\theta d\varphi, \quad (4)$$

where XPR–cross polarization power ratio, G_θ denotes elevation power gain, G_φ –azimuth power gain, P_φ , and P_θ represents the azimuth and elevation angular function of incident power spectral density. For proposed MIMO antenna the MEG difference is less than 1 dB which assures that the proposed antenna exhibits better MIMO characteristics.

6.4. Total Active Reflection Coefficient and Channel Capacity Loss. Total active reflection coefficient (TARC) and channel capacity Loss (CCL) characterize the bandwidth, frequency, and radiation capability of the MIMO antennas [30]. It can be determined using the following equations:

$$TARC = \frac{\sqrt{\sum_{i=1}^N |b_i|^2}}{\sqrt{\sum_{i=1}^N |a_i|^2}}, \quad (5)$$

$$CCL = -\log \det(\varphi^R), \quad (6)$$

where a_i and b_i are incident and reflected signals, respectively, φ^R indicates the receiver side correlation matrix. Figures 25(a) and 25(b) represent the TARC characteristics of the proposed quad port PRMA under forward and reverse biased conditions, respectively. For both cases, the TARC found to be less than –12 dB. Also, the CCL characteristics of the proposed quad port PRMA are depicted in Figures 26(a) and 26(b) under forward and reverse biased conditions. The CCL is found to be less than 0.3 bits/s/Hz which is less than the acceptable limit of 0.4 bits/s/Hz. Table 2 shows the different diversity metrics of MIMO antennas at 3.5 and 4.5 GHz for different polarization states of four ports. The diversity metrics are tabulated at the centre frequencies of 3.5 GHz and 4.7 GHz of 5G NR78 and 5G NR79 frequency bands. Hence, it is clearly noticed that the diversity metrics

of the proposed quad port PRMA is obtained well to deliver good MIMO performance. In addition, a comparative study is presented in Table 3 with few related works to highlight the salient features of the proposed antenna.

The proposed quad port PRMA is compact in size as compared to [13, 14, 16–18]. Also, it has high ARBW and impedance bandwidth than [13–18, 30]. In addition, the proposed antenna provides the flexibility to change the polarization behaviour in varying environment.

7. Conclusions

A quad port PRMA is reported for indoor environments to suppress the interferences and fading effects caused by multipath propagation. The presented antenna achieves impedance bandwidth of 50% when operating in different polarization states. The designed MIMO antenna is capable to offer RHCP, LHCP, and LP simultaneously for the band of interest 3.1 to 5.1 GHz. The proposed antenna maintains the peak gain of 4.5 dBi and efficiency of 80% over the desired operating range of frequencies. Also, the MIMO performance is validated using the metrics ECC, DG, MEG, TARC, and CCL and provides good polarization diverse nature. Hence, the proposed antenna is a potential choice for high-density varying indoor scenarios such as airports, railway stations, and shopping malls.

Data Availability

The data used to support the findings are available from the corresponding upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

Acknowledgments

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